

TRIGONOMETRY

Trigonometry Ratio

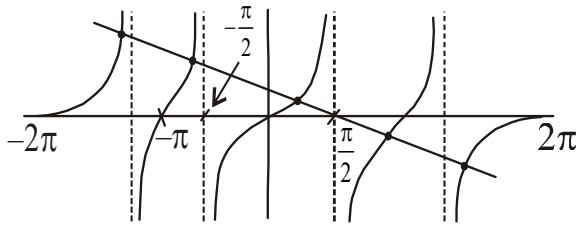
1. The number of solutions of the equation $2x + 3 \tan x = \pi$, $x \in [-2\pi, 2\pi] - \left\{ \pm \frac{\pi}{2}, \pm \frac{3\pi}{2} \right\}$ is :

समीकरण $2x + 3 \tan x = \pi$, $x \in [-2\pi, 2\pi] - \left\{ \pm \frac{\pi}{2}, \pm \frac{3\pi}{2} \right\}$ के हलों की संख्या है :

- (A) 6 (B) 5 (C) 3 (D) 4

Ans. B

Sol. $\tan x = \frac{\pi}{3} - \frac{2x}{3}$



2. If $10 \sin^4 \theta + 15 \cos^4 \theta = 6$, then the value of $\frac{27 \operatorname{cosec}^6 \theta + 8 \sec^6 \theta}{16 \sec^8 \theta}$ is :

यदि $10 \sin^4 \theta + 15 \cos^4 \theta = 6$ है, तो $\frac{27 \operatorname{cosec}^6 \theta + 8 \sec^6 \theta}{16 \sec^8 \theta}$ का मान है :

- (A) $\frac{3}{4}$ (B) $\frac{2}{5}$ (C) $\frac{1}{5}$ (D) $\frac{3}{5}$

Ans. B

Sol. $10 \sin^4 \theta + 15 \cos^4 \theta = 6$

$$\frac{27 \operatorname{cosec}^6 \theta + 8 \sec^6 \theta}{16 \sec^8 \theta} = ?$$

$$10 \tan^4 \theta + 15 = 6 \sec^4 \theta$$

$$= 6(\sec^2 \theta)^2$$

$$= 6(1 + \tan^2 \theta)^2$$

$$= 6(1 + \tan^4 \theta + 2 \tan^2 \theta)$$

$$4 \tan^4 \theta - 12 \tan^2 \theta + 9 = 0$$

$$(2 \tan^2 \theta - 3)^2 = 0$$

$$\tan^2 \theta = \frac{3}{2}$$

$$\sec^2 \theta = 1 + \tan^2 \theta$$

$$= 1 + \frac{3}{2}$$

$$\sec^2 \theta = \frac{5}{2}$$

$$\& \cot^2 \theta = \frac{2}{3}$$

$$\Rightarrow \operatorname{cosec}^2 \theta = 1 + \frac{2}{3} = \frac{5}{3}$$

$$\text{Now, } \frac{27(\operatorname{cosec}^2 \theta)^3 + 8(\sec^2 \theta)^3}{16(\sec^2 \theta)^4}$$

$$\frac{27\left(\frac{5}{3}\right)^3 + 8\left(\frac{5}{2}\right)^3}{16\left(\frac{5}{2}\right)^4}$$

$$\frac{5^3 + 5^3}{5^4} \Rightarrow \frac{2 \cdot 5^3}{5^4}$$

$$\Rightarrow \frac{2}{5}$$

3. Let α_θ and β_θ be the distinct roots of $2x^2 + (\cos \theta)x - 1 = 0$, $\theta \in (0, 2\pi)$. if m and M are the minimum and the maximum values of $\alpha_\theta^4 + \beta_\theta^4$, then $16(M + m)$ equals :

माना कि α_θ और β_θ , $2x^2 + (\cos \theta)x - 1 = 0$, $\theta \in (0, 2\pi)$ के भिन्न मूल हैं। यदि m और M , $\alpha_\theta^4 + \beta_\theta^4$ के न्यूनतम और उच्चतम मान हैं, तो $16(M + m)$ बराबर है :

(A) 17

(B) 24

(C) 27

(D) 25

Ans. D

Sol. $2x^2 + (\cos \theta)x - 1 = 0$ $\theta \in (0, 2\pi)$

$$\alpha_\theta + \beta_\theta = -\frac{\cos \theta}{2} \quad \alpha_\theta \cdot \beta_\theta = -\frac{1}{2}$$

$$\text{Now } = \alpha_\theta^4 + \beta_\theta^4$$

$$= (\alpha_\theta^2 + \beta_\theta^2)^2 - 2\alpha_\theta^2 \beta_\theta^2$$

$$= \left\{ (\alpha_\theta + \beta_\theta)^2 - 2\alpha_\theta \cdot \beta_\theta \right\}^2 - 2\alpha_\theta^2 \beta_\theta^2$$

$$= \left\{ \frac{\cos^2 \theta}{4} + 1 \right\}^2 - 2 \times \frac{1}{4}$$

$$= \left(1 + \frac{1}{4} \cos^2 \theta \right)^2 - \frac{1}{2}$$

$$[0, 1]$$

Min value when $\cos^2 \theta = 0$

$$m = \frac{1}{2}$$

Max. value when $\cos^2 \theta = 1$

$$\Rightarrow \left(1 + \frac{1}{4} \right)^2 - \frac{1}{2} = \frac{25}{16} - \frac{1}{2} = \frac{17}{16}$$

$$M = \frac{17}{16}$$

Now $16(M + m)$

$$\Rightarrow 16 \left(\frac{17}{16} + \frac{1}{2} \right) = 25$$

4. The value of $(\sin 70^\circ)(\cot 10^\circ \cot 70^\circ - 1)$ is :

$(\sin 70^\circ)(\cot 10^\circ \cot 70^\circ - 1)$ का मान है :

(A) $2/3$

(B) $3/2$

(C) 1

(D) 0

Ans. C

Sol. $\sin 70^\circ \left(\frac{\cos 10^\circ \cos 70^\circ - \sin 10^\circ \sin 70^\circ}{\sin 10^\circ \sin 70^\circ} \right)$

$$= \frac{\cos(80)}{\sin 10^\circ} = 1$$

5. If $\sum_{r=1}^{13} \left\{ \frac{1}{\sin \left(\frac{\pi}{4} + (r-1) \frac{\pi}{6} \right) \sin \left(\frac{\pi}{4} + \frac{r\pi}{6} \right)} \right\} = a\sqrt{3} + b$, $a, b \in \mathbb{Z}$, then $a^2 + b^2$ is equal to :

यदि $\sum_{r=1}^{13} \left\{ \frac{1}{\sin\left(\frac{\pi}{4} + (r-1)\frac{\pi}{6}\right) \sin\left(\frac{\pi}{4} + \frac{r\pi}{6}\right)} \right\} = a\sqrt{3} + b$, $a, b \in \mathbb{Z}$, तो $a^2 + b^2$ है, तो बराबर है :

(A) 2

(B) 4

(C) 8

(D) 10

Ans. C

Sol.
$$\sum_{r=1}^{13} \left[\frac{1}{\sin\left(\frac{\pi}{4} + (r-1)\frac{\pi}{6}\right) \sin\left(\frac{\pi}{4} + \frac{r\pi}{6}\right)} \right]$$

$$\frac{1}{\left(\sin\left(\frac{\pi}{6}\right)\right)} \sum_{r=1}^{13} \left[\frac{\sin\left(\left(\frac{\pi}{4} + \frac{r\pi}{6}\right) - \left(\frac{\pi}{4} + \frac{(r-1)\pi}{6}\right)\right)}{\sin\left(\frac{\pi}{4} + (r-1)\frac{\pi}{6}\right) \sin\left(\frac{\pi}{4} + \frac{r\pi}{6}\right)} \right]$$

$$\frac{1}{\sin\left(\frac{\pi}{6}\right)} \sum_{r=1}^{13} \left[\cot\left(\frac{\pi}{4} + \frac{(r-1)\pi}{6}\right) - \cot\left(\frac{\pi}{4} + \frac{r\pi}{6}\right) \right]$$

$$(2) \left[\left(\cot\left(\frac{\pi}{4}\right) - \cot\left(\frac{\pi}{4} + \frac{\pi}{6}\right) \right) + \left(\cot\left(\frac{\pi}{4} + \frac{\pi}{6}\right) - \cot\left(\frac{\pi}{4} + \frac{2\pi}{6}\right) \right) + \dots + \cot\left(\frac{\pi}{4} + \frac{12\pi}{6}\right) - \cot\left(\frac{\pi}{4} + \frac{13\pi}{6}\right) \right]$$

$$(2) \left[\cot\left(\frac{\pi}{4}\right) - \cot\left(\frac{\pi}{4} + 2\pi + \frac{\pi}{6}\right) \right]$$

$$(B) [1 - \cot(75^\circ)]$$

$$(2) (1 - (2 - \sqrt{3}))$$

$$(2) (\sqrt{3} - 1)$$

$$(2\sqrt{3} - 2) \Rightarrow a\sqrt{3} + b \Rightarrow a = 2, b = -2$$

$$a^2 + b^2 = 4 + 4 = 8$$

6. If $\sin x + \sin^2 x = 1$, $x \in \left(0, \frac{\pi}{2}\right)$, then

$(\cos^{12} x + \tan^{12} x) + 3(\cos^{10} x + \tan^{10} x + \cos^8 x + \tan^8 x) + (\cos^6 x + \tan^6 x)$ is equal to :

यदि $\sin x + \sin^2 x = 1, x \in \left(0, \frac{\pi}{2}\right)$ है, तो

$(\cos^{12} x + \tan^{12} x) + 3(\cos^{10} x + \tan^{10} x + \cos^8 x + \tan^8 x) + (\cos^6 x + \tan^6 x)$ बराबर है :

- (A) 4 (B) 2 (C) 3 (D) 1

Ans. B

Sol. $\sin x + \sin^2 x = 1$

$$\Rightarrow \sin x = \cos^2 x \quad \Rightarrow \tan x = \cos x$$

$\therefore$ Given expression

$$= 2\cos^{12} x + 6[\cos^{10} x + \cos^8 x] + 2\cos^6 x$$

$$= 2[\sin^6 x + 3\sin^5 x + 3\sin^4 x + \sin^3 x]$$

$$= 2\sin^3 x[(\sin x + 1)^3]$$

$$= 2[\sin^2 x + \sin x]^3$$

$$= 2$$

7. Let $|\cos \theta \cdot \cos(60 - \theta) \cdot \cos(60 + \theta)| \leq \frac{1}{8}, \theta \in [0, 2\pi]$. Then, then sum of all $\theta \in [0, 2\pi]$, where $\cos 3\theta$ attains its maximum value of is :

माना $|\cos \theta \cdot \cos(60 - \theta) \cdot \cos(60 + \theta)| \leq \frac{1}{8}, \theta \in [0, 2\pi]$ है। तो $\theta \in [0, 2\pi]$ के सभी मानों, जिनके लिए $\cos 3\theta$ का मान

उच्चतम है, का योग है :

- (A) 18π (B) 9π (C) 15π (D) 6π

Ans. D

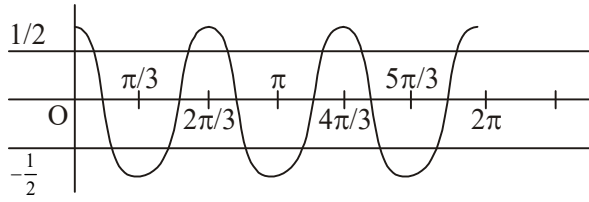
Sol. $\cos \theta \cdot \cos(60 - \theta) \cos(60 + \theta) = \frac{1}{4} \cos 3\theta$

$$\therefore |\cos 3\theta| \leq \frac{1}{2}$$

$$0 \leq 3\theta \leq 6\pi$$

$$-\frac{1}{2} \leq \cos 3\theta \leq \frac{1}{2}$$

$$\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{9\pi}{6}, \frac{11\pi}{6}$$



θ satisfying condition are $\frac{\pi}{9}, \frac{\pi}{2} + \frac{\pi}{18} + \frac{5\pi}{6} - \frac{\pi}{18}, \frac{7\pi}{6} + \frac{\pi}{16}, \frac{3\pi}{2} - \frac{\pi}{18}, \frac{11\pi}{6} + \frac{\pi}{18}$

$$\begin{aligned}\sum \theta &= \frac{\pi}{9} + \frac{\pi}{2} + \frac{\pi}{6} + \frac{7\pi}{6} + \frac{3\pi}{2} + \frac{11\pi}{6} + \frac{\pi}{18} \\ &= \frac{(2+9+15+21+27+33+1)\pi}{18} = 6\pi\end{aligned}$$

8. If $\tan A = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan B = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and $\tan C = (x^{-3} + x^{-2} + x^{-1})^{\frac{1}{2}}$, $0 < A, B, C < \frac{\pi}{2}$, then $A + B$ is equal to :

यदि $\tan A = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan B = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ तथा $\tan C = (x^{-3} + x^{-2} + x^{-1})^{\frac{1}{2}}$, $0 < A, B, C < \frac{\pi}{2}$ हैं, तो

$A + B$ बराबर है :

- (A) $\pi - C$ (B) $\frac{\pi}{2} - C$ (C) $2\pi - C$ (D) C

Ans. D

Sol. $\tan A = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan B = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

$$\tan(A+B) = \frac{\frac{1}{\sqrt{x(x^2+x+1)}} + \frac{\sqrt{x}}{\sqrt{x^2+x+1}}}{1 - \frac{1}{\sqrt{x(x^2+x+1)}} \cdot \frac{\sqrt{x}}{\sqrt{x^2+x+1}}}$$

$$\tan(A+B) = \frac{1+x}{\sqrt{x} \cdot \sqrt{x^2+x+1}} \times \frac{(x^2+x+1)}{x^2+x}$$

$$\tan(A+B) = \frac{\sqrt{x^2+x+1}}{x\sqrt{x}} \quad \dots(1)$$

$$\text{Given } \tan C = \left(\frac{1}{x^3} + \frac{1}{x^2} + \frac{1}{x} \right)^{\frac{1}{2}}$$

$$\tan C = \frac{\sqrt{1+x+x^2}}{x\sqrt{x}} \quad \dots(2)$$

from equation (1) & (2)

$$\tan(A+B) = \tan C$$

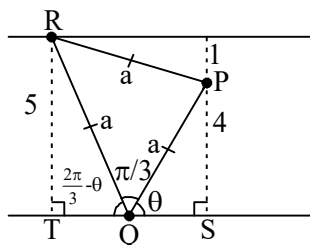
$$\text{or } A+B = C$$

9. Let the distance between two parallel lines be 5 units and a point P lie between the lines at a unit distance from one of them. An equilateral triangle PQR is formed such that Q lies on one of the parallel lines, while R lies on the other. Then $(QR)^2$ is equal to _____.

माना कि दो समान्तर रेखाओं के बीच की दूरी 5 इकाई है और एक बिन्दु P रेखाओं के बीच उनमें से एक से 1 इकाई की दूरी पर स्थित है। एक समबाहु त्रिभुज PQR इस प्रकार बनाया जाता है कि Q समान्तर रेखाओं में से एक पर जबकि R दूसरी पर स्थित हैं तब $(QR)^2$ बराबर है _____।

Ans. 28

Sol.



$$\text{In } \triangle PQS \quad \sin \theta = \frac{4}{a}$$

$$\text{In } \triangle QRT \quad \sin \left(\frac{2\pi}{3} - \theta \right) = \frac{5}{a}$$

$$\sin \frac{2\pi}{3} \cos \theta - \cos \frac{2\pi}{3} \sin \theta = \frac{5}{a}$$

$$\frac{\sqrt{3}}{2} \times \frac{\sqrt{a^2-16}}{a} + \frac{1}{2} \times \frac{4}{a} = \frac{5}{a}$$

$$\sqrt{3}\sqrt{a^2-16} + 4 = 10$$

$$a^2 - 16 = 12$$

$$a^2 = 28$$

$$(QR)^2 = 28$$

10. If the range of $f(\theta) = \frac{\sin^4 \theta + 3 \cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$, $\theta \in \mathbb{R}$ is $[\alpha, \beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to _____.

यदि $f(\theta) = \frac{\sin^4 \theta + 3 \cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$, $\theta \in \mathbb{R}$ का परिसर $[\alpha, \beta]$ है, तब अनंत गुणोत्तर श्रेणी का योग क्या होगा, जिसका प्रथम पद 64

तथा सार्वनुपात $\frac{\alpha}{\beta}$ है, _____

Ans. 96

Sol. $f(\theta) = \frac{\sin^4 \theta + 3 \cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$

$$f(\theta) = 1 + \frac{2 \cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$$

$$f(\theta) = \frac{2 \cos^2 \theta}{\cos^4 \theta - \cos^2 \theta + 1} + 1$$

$$f(\theta) = \frac{2}{\cos^2 \theta + \sec^2 \theta - 1} + 1$$

$$f(\theta)|_{\min} = 1$$

$$f(\theta)|_{\max} = 3$$

$$S = \frac{64}{1 - 1/3} = 96$$

Trigonometry Equation

11. If $\theta \in [-2\pi, 2\pi]$, then the number of solutions of $2\sqrt{2} \cos^2 \theta + (2 - \sqrt{6}) \cos \theta - \sqrt{3} = 0$, is equal to :

यदि $\theta \in [-2\pi, 2\pi]$ है, तो समीकरण $2\sqrt{2} \cos^2 \theta + (2 - \sqrt{6}) \cos \theta - \sqrt{3} = 0$ के हलों की संख्या है :

(A) 6

(B) 10

(C) 12

(D) 8

Ans. D

Sol. $2\sqrt{2} \cos^2 \theta + 2 \cos \theta - \sqrt{6} \cos \theta - \sqrt{3} = 0$

$$(2 \cos \theta - \sqrt{3})(\sqrt{2} \cos \theta + 1) = 0$$

$$\cos \theta = \frac{\sqrt{3}}{2}, -\frac{1}{\sqrt{2}}$$

Number of solution = 8

12. The number of solutions of the equation $(4 - \sqrt{3}) \sin x - 2\sqrt{3} \cos^2 x = -\frac{4}{1 + \sqrt{3}}$, $x \in \left[-2\pi, \frac{5\pi}{2}\right]$ is :

समीकरण $(4 - \sqrt{3}) \sin x - 2\sqrt{3} \cos^2 x = -\frac{4}{1 + \sqrt{3}}$, $x \in \left[-2\pi, \frac{5\pi}{2}\right]$ के हलों की संख्या है :

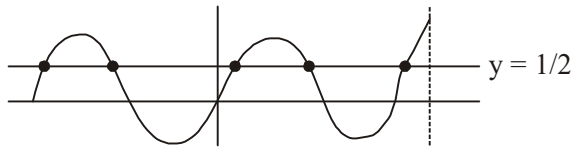
- (A) 6 (B) 5 (C) 4 (D) 3

Ans. B

Sol. $(4 - \sqrt{3}) \sin x - 2\sqrt{3} (1 - \sin^2 x) = 2(1 - \sqrt{3})$

$$\Rightarrow (2 \sin x - 1)(\sqrt{3} \sin x + 2) = 0$$

$$\Rightarrow \sin x = \frac{1}{2}$$



So number of solutions = 5

13. The number of solutions of the equation $\cos 2\theta \cos \frac{\theta}{2} + \cos \frac{5\theta}{2} = 2 \cos^3 \frac{5\theta}{2}$ in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is :

समीकरण $\cos 2\theta \cos \frac{\theta}{2} + \cos \frac{5\theta}{2} = 2 \cos^3 \frac{5\theta}{2}$ के $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ में हलों की संख्या है :

- (A) 5 (B) 7 (C) 9 (D) 6

Ans. B

Sol. $\cos 2\theta \cdot \cos \frac{\theta}{2} = \cos \frac{5\theta}{2} \left(2 \cos^2 \frac{5\theta}{2} - 1 \right)$

$$\Rightarrow 2 \cos 2\theta \cdot \cos \frac{\theta}{2} = 2 \cos \frac{5\theta}{2} \cdot \cos 5\theta$$

$$\Rightarrow \cos \frac{5\theta}{2} + \cos \frac{3\theta}{2} = \cos \frac{15\theta}{2} + \cos \frac{5\theta}{2}$$

$$\Rightarrow \cos \frac{15\theta}{2} - \cos \frac{3\theta}{2} = 0$$

$$\Rightarrow 2 \sin\left(\frac{9\theta}{2}\right) \sin(3\theta) = 0$$

$$\text{Either } \sin\left(\frac{9\theta}{2}\right) = 0 \Rightarrow \frac{9\theta}{2} = n\pi$$

$$\Rightarrow \theta = \frac{2n\pi}{9} \Rightarrow \theta = -\frac{4\pi}{9}, -\frac{2\pi}{9}, 0, \frac{2\pi}{9}, \frac{4\pi}{9}$$

$$\text{or } \sin(3\theta) = 0$$

$$\Rightarrow 3\theta = n\pi$$

$$\Rightarrow \theta = \frac{n\pi}{3}$$

$$\Rightarrow \theta = -\frac{\pi}{3}, 0, \frac{\pi}{3}$$

So, total solutions = 7

14. Let $A = \left\{ x \in (0, \pi) - \left\{ \frac{\pi}{2} \right\} : \log_{(2/\pi)} |\sin x| + \log_{(2/\pi)} |\cos x| = 2 \right\}$ and

$$B = \left\{ x \geq 0 : \sqrt{x}(\sqrt{x} - 4) - 3|\sqrt{x} - 2| + 6 = 0 \right\}. \text{ Then } n(A \cup B) \text{ is equal to :}$$

माना कि $A = \left\{ x \in (0, \pi) - \left\{ \frac{\pi}{2} \right\} : \log_{(2/\pi)} |\sin x| + \log_{(2/\pi)} |\cos x| = 2 \right\}$ और

$$B = \left\{ x \geq 0 : \sqrt{x}(\sqrt{x} - 4) - 3|\sqrt{x} - 2| + 6 = 0 \right\} \text{ है। तब } n(A \cup B) \text{ बराबर है :}$$

(A) 4

(B) 2

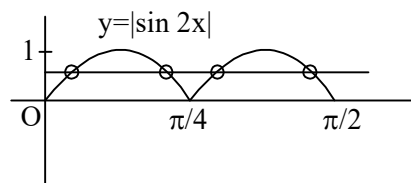
(C) 6

(D) 8

Ans. D

Sol. $A : |\sin x \cos x| = \left(\frac{2}{\pi}\right)^2$

$$|\sin 2x| = \frac{8}{\pi^2}$$



(4 solutions)

B : Put $\sqrt{x} - 2 = t$

$$(t^2 - 4) - 3|t| + 6 = 0$$

$$|t|^2 - 3|t| + 2 = 0$$

$$|t| = 1, 2$$

$$x = 1, 9; 0, 16 \text{ (4 solutions)}$$

15. Suppose $\theta \in \left[0, \frac{\pi}{4}\right]$ is a solutions of $4 \cos \theta - 3 \sin \theta = 1$. Then $\cos \theta$ is equal to :

माना $4 \cos \theta - 3 \sin \theta = 1$ का एक हल $\theta \in \left[0, \frac{\pi}{4}\right]$ है। तो $\cos \theta$ बराबर है :

(A) $\frac{4}{(3\sqrt{6}+2)}$ (B) $\frac{4}{(3\sqrt{6}-2)}$ (C) $\frac{6-\sqrt{6}}{(3\sqrt{6}-2)}$ (D) $\frac{6+\sqrt{6}}{(3\sqrt{6}+2)}$

Ans. B

Sol. $4 \cos \theta - 3 \sin \theta = 1$

$$4 \cos \theta - 1 = 3 \sin \theta$$

$$16 \cos^2 \theta + 1 - 8 \cos \theta = 9(1 - \cos^2 \theta)$$

$$\Rightarrow 25 \cos^2 \theta - 8 \cos \theta - 8 = 0$$

$$\Rightarrow \cos \theta = \frac{8 \pm \sqrt{64 + 4 \times 25 \times 8}}{2.25}$$

$$= \frac{8 \pm 4\sqrt{4+50}}{2.25}$$

$$= \frac{4 \pm 2\sqrt{54}}{25}$$

As $\theta \in \left[0, \frac{\pi}{4}\right]$

$$\Rightarrow \cos \theta = \frac{4+6\sqrt{6}}{25} = \frac{4}{3\sqrt{6}-2}$$

16. If $2 \tan^2 \theta - 5 \sec \theta = 1$ has exactly 7 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, for the least value of $n \in \mathbb{N}$, then $\sum_{k=1}^n \frac{k}{2^k}$ is equal to :

जिस $n \in \mathbb{N}$ के न्यूनतम मान के लिए अंतराल $\left[0, \frac{n\pi}{2}\right]$ में $2 \tan^2 \theta - 5 \sec \theta = 1$ के ठीक 7 हल हैं, उस n के लिए $\sum_{k=1}^n \frac{k}{2^k}$ बराबर है :s

(A) $\frac{1}{2^{14}}(2^{15}-15)$ (B) $1-\frac{15}{2^{13}}$ (C) $\frac{1}{2^{13}}(2^{14}-15)$ (D) $\frac{1}{2^{15}}(2^{14}-14)$

Ans. C

Sol. $2 \tan^2 \theta - 5 \sec \theta - 1 = 0$

$$\Rightarrow 2 \sec^2 \theta - 5 \sec \theta - 3 = 0$$

$$\Rightarrow (2 \sec \theta + 1)(\sec \theta - 3) = 0$$

$$\Rightarrow \sec \theta = -\frac{1}{2}, 3$$

$$\Rightarrow \cos \theta = -2, \frac{1}{3}$$

$$\Rightarrow \cos \theta = \frac{1}{3}$$

For 7 solutions $n = 13$

So, $\sum_{k=1}^{13} \frac{k}{2^k} = S$ (say)

$$S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{13}{2^{13}}$$

$$\frac{1}{2}S = \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{12}{2^{13}} + \frac{13}{2^{14}}$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} \cdot \frac{1 - \frac{1}{2^{13}}}{1 - \frac{1}{2}} - \frac{13}{2^{14}}$$

$$\Rightarrow S = 2 \cdot \left(\frac{2^{13}-1}{2^{13}} \right) - \frac{13}{2^{13}}$$

17. If $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is the solution of $4 \cos \theta + 5 \sin \theta = 1$, then the value of $\tan \alpha$ is :

यदि समीकरण $4 \cos \theta + 5 \sin \theta = 1$ का हल $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ है, तो $\tan \alpha$ का मान है :

(A) $\frac{10-\sqrt{10}}{12}$ (B) $\frac{\sqrt{10}-10}{6}$ (C) $\frac{10-\sqrt{10}}{6}$ (D) $\frac{\sqrt{10}-10}{12}$

Ans. D

Sol. $4 + 5 \tan \theta = \sec \theta$

Squaring : $24 \tan^2 \theta + 40 \tan \theta + 15 = 0$

$$\tan \theta = \frac{-10 \pm \sqrt{10}}{12}$$

and $\tan \theta = -\left(\frac{10 + \sqrt{10}}{12}\right)$ is Rejected

18. The sum of the solutions $x \in \mathbb{R}$ of the equation $\frac{3 \cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$ is :

समीकरण $\frac{3 \cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$ के हल $x \in \mathbb{R}$ का योगफल है :

(A) 1 (B) 0 (C) -1 (D) 3

Ans. A

Sol. Given $\frac{3 \cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$

$$\Rightarrow \frac{\cos 2x (3 + \cos^2 2x)}{(\cos^2 x - \sin^2 x) [\sin^4 x + \cos^4 x + \sin^2 x \cos^2 x]}$$

$$= x^3 - x^2 + 6$$

$$\Rightarrow \frac{3 + \cos^2 2x}{1 - \sin^2 x \cos^2 x} = x^3 - x^2 + 6$$

$$\Rightarrow 4 \left(\frac{3 + \cos^2 2x}{4 - \sin^2 2x} \right) = x^3 - x^2 + 6$$

$$\Rightarrow x^3 - x^2 + 2 = 0$$

$$\Rightarrow (x+1)(x^2 - 2x + 2) = 0$$

$$\Rightarrow (x+1)((x-1)^2 + 1) = 0$$

$\therefore$ Sum of solutions = -1

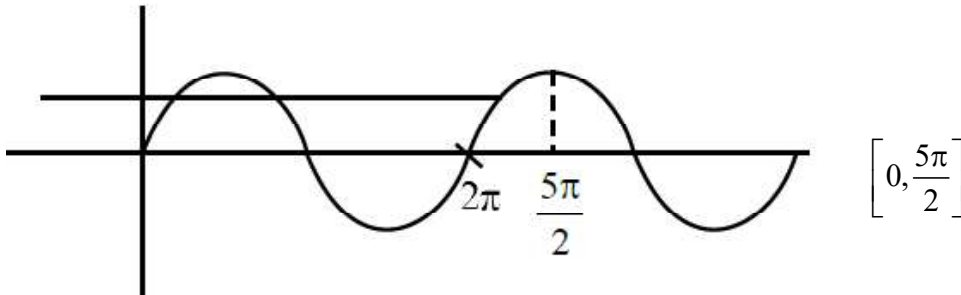
19. If $2\sin^3 x + \sin 2x \cos x + 4 \sin x - 4 = 0$ has exactly 3 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, $n \in \mathbb{N}$, then the roots of the equation $x^2 + nx + (n-3) = 0$ belongs to :

यदि $2\sin^3 x + \sin 2x \cos x + 4 \sin x - 4 = 0$ के अंतराल $\left[0, \frac{n\pi}{2}\right]$, $n \in \mathbb{N}$ में ठीक 3 हल हैं, तो समीकरण $x^2 + nx + (n-3) = 0$ के मूल किस में हैं?

- (A) $(-\infty, 0)$ (B) $\left(-\frac{\sqrt{17}}{2}, \frac{\sqrt{17}}{2}\right)$ (C) $(0, \infty)$ (D) $\mathbb{Z}$

Ans. A

Sol. $2\sin^3 x + 2 \sin x \cos^2 x + 4 \sin x - 4 = 0$
 $\Rightarrow 2\sin^3 x + 2\sin x (1 - \sin^2 x) + 4 \sin x - 4 = 0$
 $\Rightarrow 2\sin^3 x + 6 \sin x - 2 \sin^3 x - 4 = 0$
 $\sin x = \frac{2}{3}$



$$x^2 + 5x + 2 = 0$$

$$\alpha, \beta = \frac{-5 \pm \sqrt{17}}{2} \text{ both negative, hence } (-\infty, 0)$$

20. The number of solutions of the equation $4\sin^2 x - 4\cos^3 x + 9 - 4\cos x = 0$; $x \in [-2\pi, 2\pi]$ is :

समीकरण $4\sin^2 x - 4\cos^3 x + 9 - 4\cos x = 0$; $x \in [-2\pi, 2\pi]$ के हलों की संख्या है :

- (A) 0 (B) 3 (C) 1 (D) 2

Ans. A

Sol. $4 - 4\cos^2 x - 4\cos^3 x + 9 - 4\cos x = 0$
 $4\cos^3 x + 4\cos^2 x + 4\cos x = 13$
 $\therefore \text{LHS} \leq 12$
 So equation will have no solution.